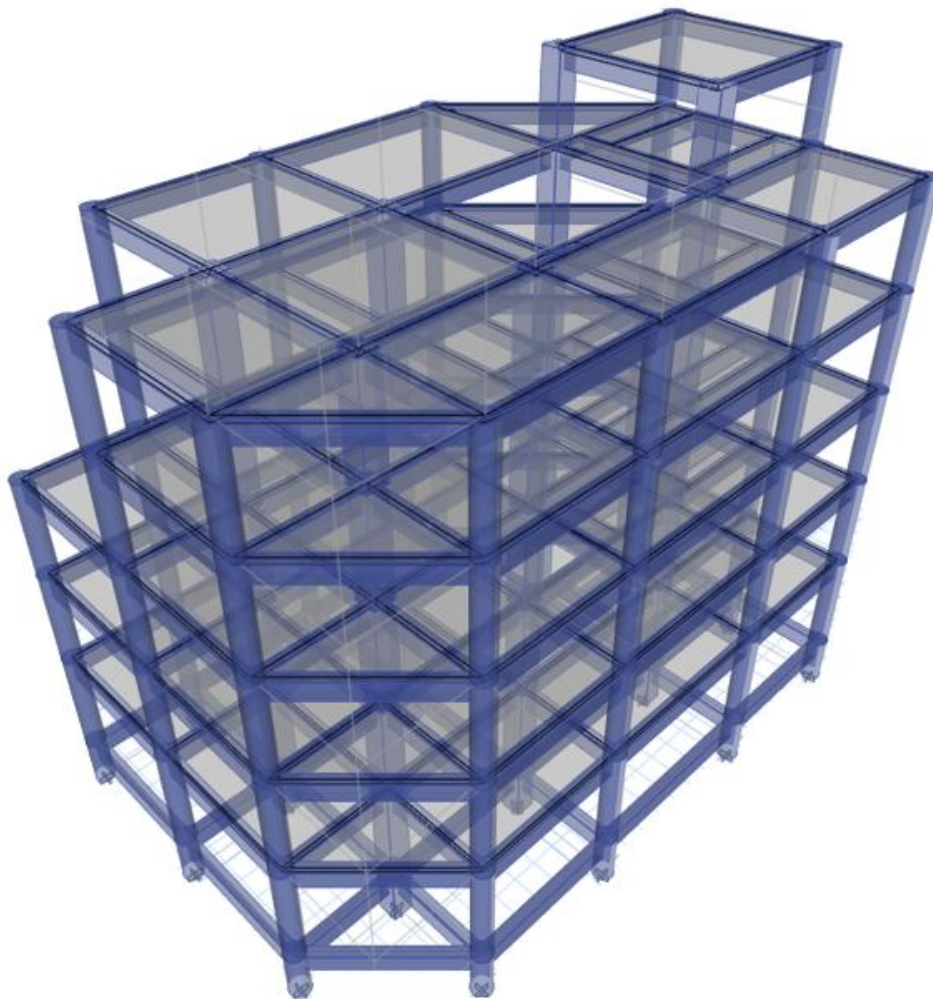


Structural Analysis and Design Report

5-Storey RC Office Building with Roof Deck

Upper General Luna Road, Baguio City



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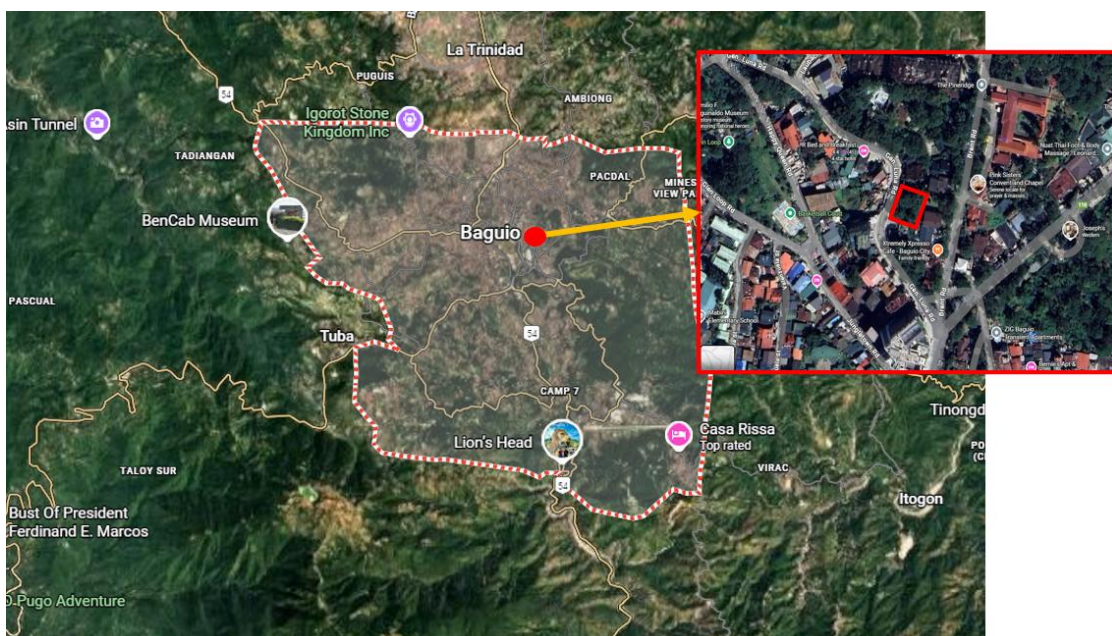
1. Introduction

The proposed project involves the construction of a Five-Storey Reinforced Concrete Office Building with Roof Deck, designed with emphasis on structural safety, efficiency, and durability. The structure will employ reinforced concrete as the primary material, ensuring strength and stability under various load conditions typical of the Baguio City environment.

The foundation system will consist of isolated footings, footing tie beams, and reinforced concrete columns, supporting a slab-on-grade at the ground floor. The superstructure will comprise reinforced concrete beams, slabs, and columns, designed to resist vertical and lateral loads in accordance with the National Structural Code of the Philippines (NSCP 2015).

To enhance the building's performance and reliability, computer-aided structural analysis software (ETABS) will be utilized to model and evaluate the behavior of the structure under dead, live, wind, and seismic loads. The analysis results will guide the proper sizing and reinforcement of structural members to ensure code compliance and safety.

The project site is located at **Upper General Luna Road, Baguio City, Benguet**, a region characterized by hilly terrain and moderate seismic activity, making structural integrity and stability essential considerations in the overall design.



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2. Design Criteria

The structural design of the building components was carried out in accordance with the provisions of the **National Structural Code of the Philippines (NSCP 2015)**. The materials and design parameters were selected to ensure the structural safety, serviceability, and durability of the building. Reinforced concrete members, such as columns, beams, slabs, and footings, were designed using the **Ultimate Strength Design Method**. The design strengths of concrete and reinforcing steel are presented in the following tables, along with the corresponding load considerations for each structural element.

Structural Elements

Members	Materials design stresses
Column	$F'_c = 27.6 \text{ Mpa (4000 psi)}$
Beams and reinforced concrete slab	$F'_c = 20.7 \text{ Mpa (3000 psi)}$
Footing	$F'_c = 20.7 \text{ Mpa (3000 psi)}$

Deformed Rebars

Deformed rebars	Grade
12 mm diameter	$F_y = 275 \text{ MPa (Grade 40 ksi)}$
16 mm diameter	$F_y = 414 \text{ MPa (Grade 60 ksi)}$

Static Seismic load	
Seismic Load Code	UBC(1997) / UBC 97
Soil Profile Type	B
Seismic Zone Factor	S_d
Seismic Source Type	4 (0.4)
Closest Distance to Known seismic Source	7.8km
Importance Factor:	1
Ductility Coefficient (R):	8.5 (SMRF)

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Static Load Case			
Load	Type	Multiplier (self w)	Auto Lateral load
D	Dead Load	1	
SD	Superimposed Dead Load	0	
RLL	Redusable Live Load	0	
L	Live Load	0	
EXstat	Seismic	0	UBC 97
EYstat	Seismic	0	UBC 97
WX	Wind	0	ASCE 7-10
WY	Wind	0	ASCE 7-10
LR	Roof Live Load	0	

Wind Load (ASCE 7-10)	
Basic Wind Speed	260kph
Exposure Category	B
Directional Factor	0.85
Gust Factor	0.85

Stiffness Scale Factors	
Member and Condition	Moment of Inertia
Columns	0.7Ig
Beams	0.35Ig

3. Design Codes

Applicable codes

NSCP 2015 7th EDITION – National Structural Code of the Philippines 2015 referenced from American Concrete Institute (Metric) ACI 318-14M for reinforced concrete design
Uniform Building Code UBC 1997 for earthquake calculation procedures American Society of Civil Engineers ASCE 7-10 for wind calculation procedures

Purpose of the code

ACI 318-14M Section 1.3

1.3 — Purpose

1.3.1 The purpose of this Code is to provide for public health and safety by establishing minimum requirements for strength, stability, serviceability, durability, and integrity of concrete structures.

NSCP 2015 Section 101.2

101.2 Purpose

The purpose of this code is to provide minimum requirements for the design of buildings, towers, and other vertical structures, and minimum standards and guidelines to safeguard life or limb, property and public welfare by regulating and controlling the design, construction, and quality of materials pertaining to the structural aspects of all buildings and structures within this jurisdiction.

4. Design Requirements

NSCP Section 104

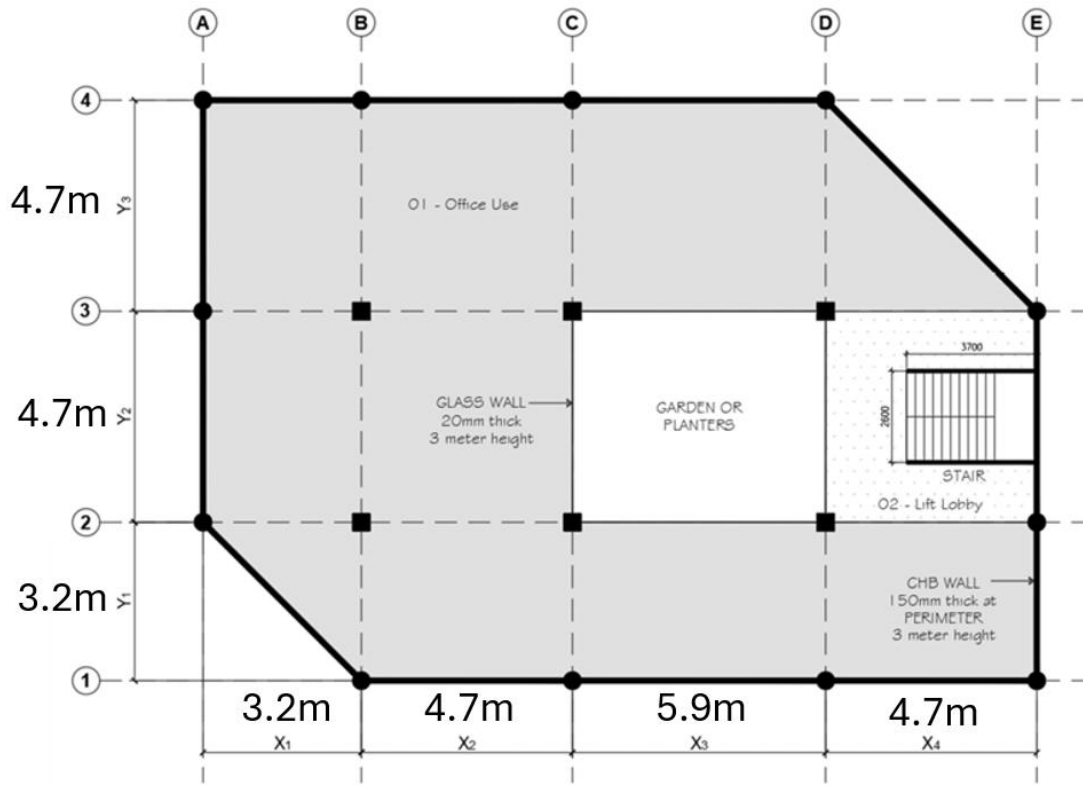
104.1 Strength Requirement

Design shall be in accordance with Strength Design, Load and Resistance Factor Design and Allowable Strength Design methods, as permitted by the applicable material chapters.

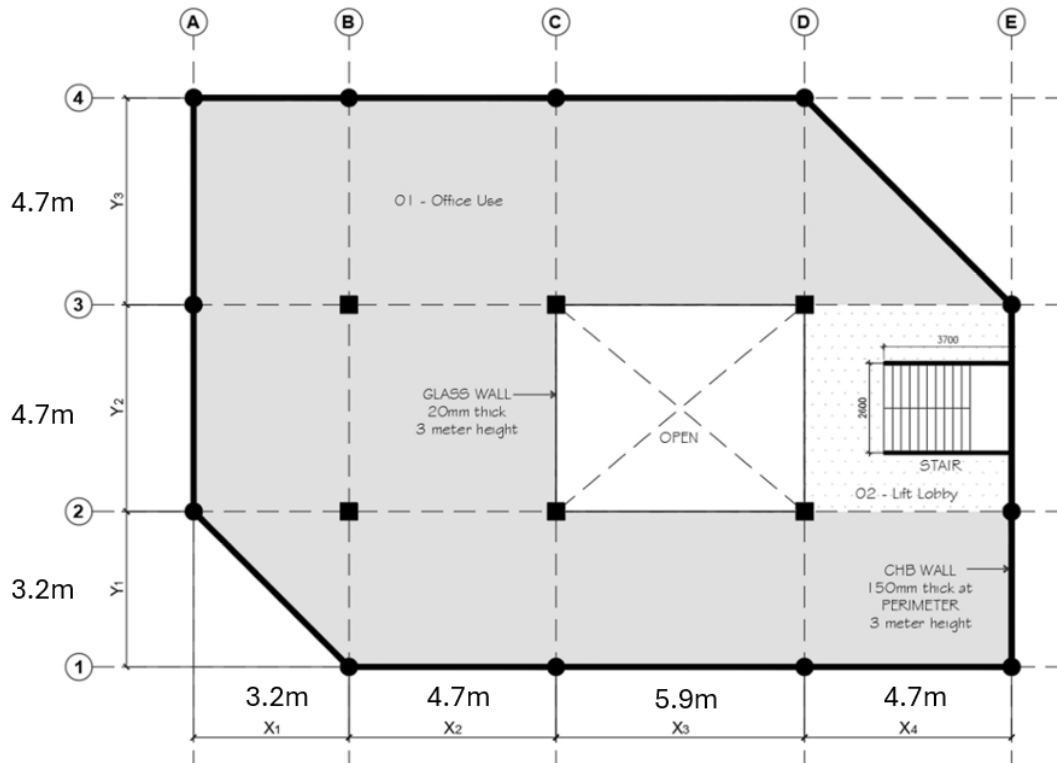
104.2 Serviceability Requirements

Structural systems and members thereof shall be designed to have adequate stiffness to limit deflections, lateral drifts, vibration, or any other deformations that adversely affect the intended use and performance of buildings, towers and other vertical structures. The design shall also consider durability, resistance to exposure to weather or aggressive environment, crack control, and other conditions that affect the use and performance of buildings, towers and other vertical structures.

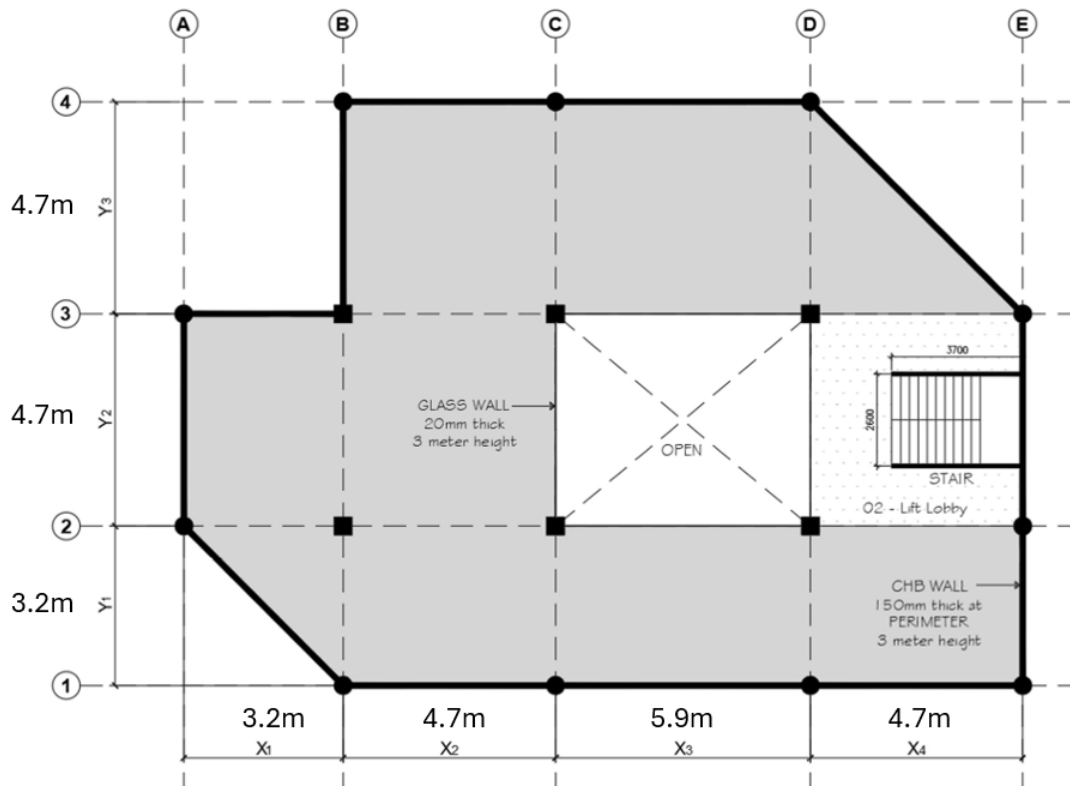
5. Floor Framing Plans



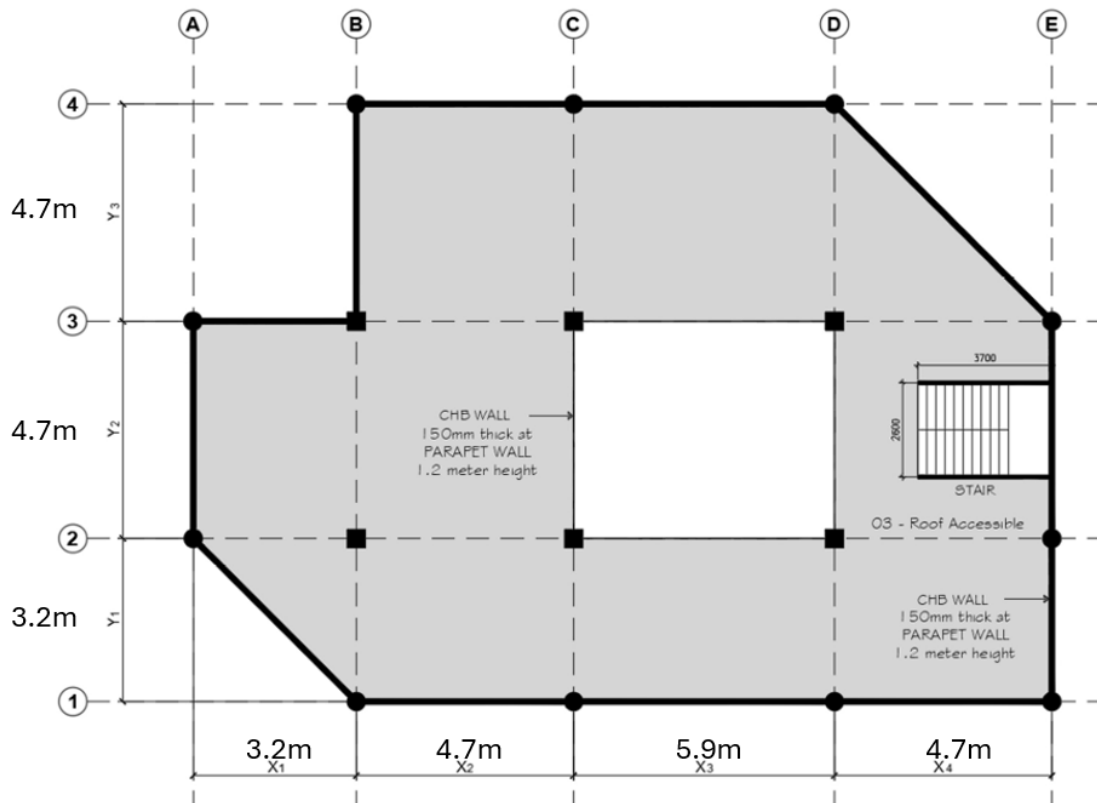
GROUND FLOOR FRAMING PLAN



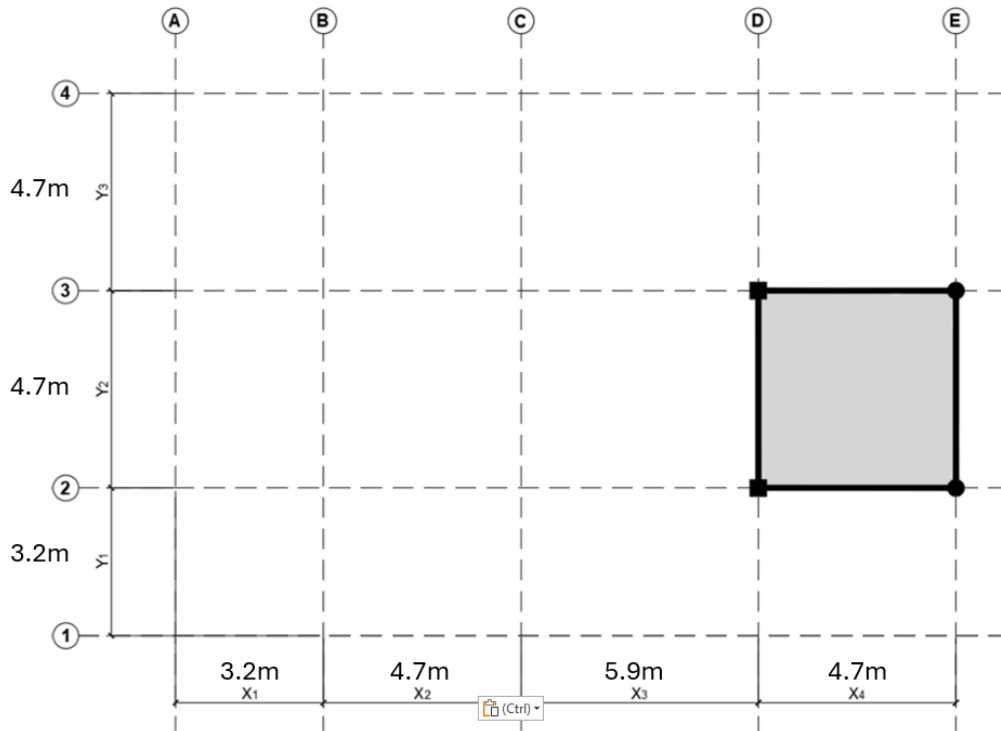
2F - 4F TYPICAL FRAMING PLAN



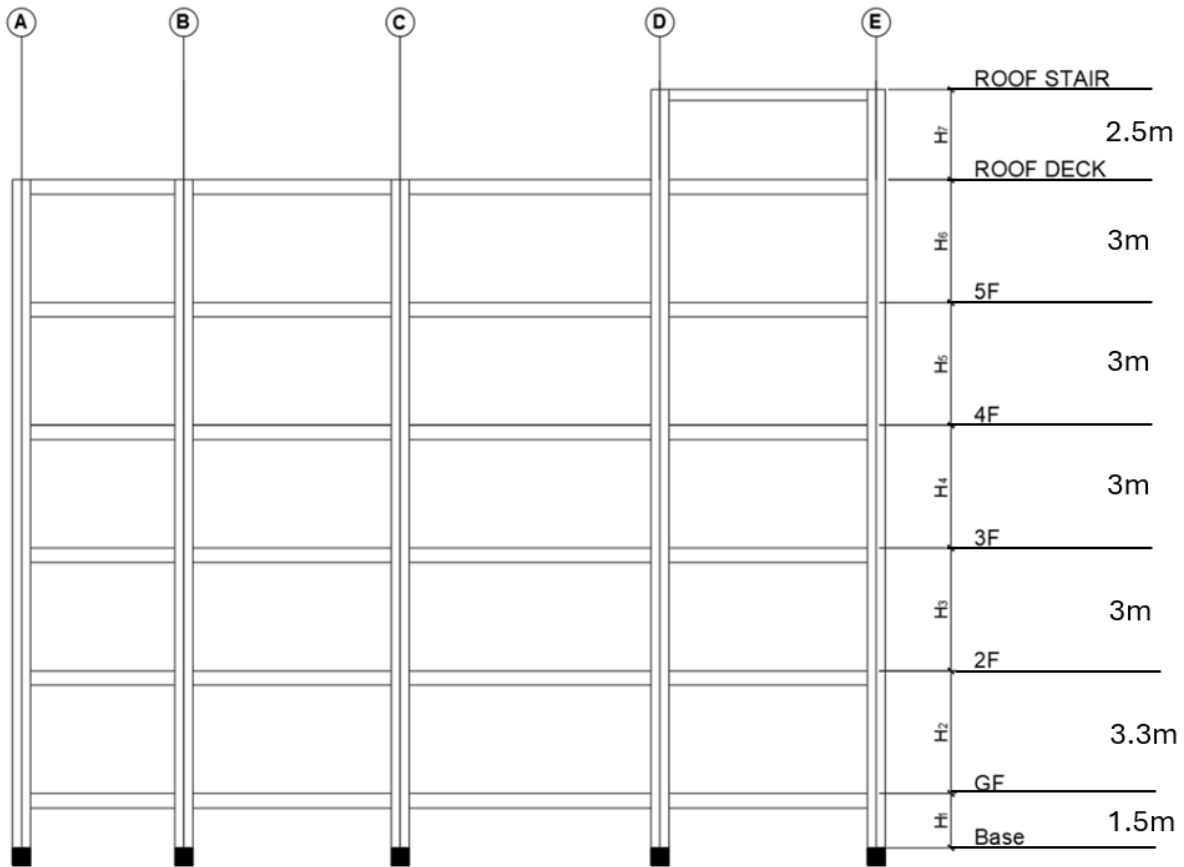
5F FRAMING PLAN



ROOF DECK FRAMING PLAN



ROOF STAIR FRAMING PLAN



ELEVATION

6. Combination of Loads

NSCP Section 203

Buildings, towers and other vertical structures and all portions thereof shall be designed to resist the load combinations specified in Section 203.3, 203.4 and 203.5.

The most critical effect can occur when one or more of the contributing loads are not acting. All applicable loads shall be considered, including both earthquake and wind, in accordance with the specified load combinations.

D = dead load

E = earthquake load set forth in Section 208.6.1

Em = estimated maximum earthquake force that can be developed in the structure as set forth in Section 208.6.1

F = loads due to fluids with well-defined pressures and maximum heights

H = load due to lateral pressure of soil and water in soil

L = live load, except roof live load, including any permitted live load reduction

Lr = roof live load, including any permitted live load reduction

P = Ponding load

R = Rain load on the undeflected roof

T = self-straining force and effects arising from contraction or expansion resulting from temperature change, shrinkage, moisture change, creep in component materials, movement due to differential settlement, or combination thereof

W = load due to wind pressure

Refer to NSCP Section 203.3

Load combinations using **Strength Design** or **Load and Resistance Factor Design**

203.3.1 Basic Load Combinations

$1.4(D + F)$	203-1
$1.2(D + F + T) + 1.6(L + H) + 0.5(L_r \text{ or } R)$	203-2
$1.2D + 1.6(L_r \text{ or } R) + (f_1L \text{ or } 0.5W)$	203-3
$1.2D + 1.0W + f_1L + 0.5(L_r \text{ or } R)$	203-4
$1.2D + 1.0E + f_1L$	203-5

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$0.9D + 1.0W + 1.6H$ 203-6

$0.9D + 1.0E + 1.6H$ 203-7

Where $f_1 = 1.0$ for floors in places of public assembly, for live loads in excess of 4.8kPa, and for garage live load, or 0.5 for other live loads

NSCP Section 203.4

Load combinations Using Allowable Stress or Allowable Strength Design

203.4.1 Basic Load Combinations

$D + F$ 203-8

$D + H + F + L + T$ 203-9

$D + H + F + (L_r \text{ or } R)$ 203-10

$D + H + F + 0.75(L + T + (L_r \text{ or } R))$ 203-11

$D + H + F + (0.6W \text{ or } E/1.4)$ 203-12

No increase in allowable stresses shall be used with these load combinations except as specifically permitted by Section 203.4.2.

203.4.2 Alternate Basic Load Combinations

In lieu of the basic load combinations specified in section 203.4.1, structures and portions thereof shall be permitted to be designed for the most critical effects resulting from the following load combinations. When using these alternate basic load combinations, a one-third increase shall be permitted in allowable stresses for all combinations, including W or E.

$D + H + F + 0.75(L + L_r + (0.6W \text{ or } E/1.4))$ 203-13

$0.6D + 0.6W + H$ 203-14

$0.6D + E/1.4 + H$ 203-15

$D + L + (L_r \text{ or } R)$ 203-16

$D + L + 0.6W$ 203-17

$D + L + E/1.4$ 203-18

203.5 Special Seismic Load Combinations

For both allowable stress design and strength design for concrete, and Load and Resistance factor Design (LRFD) and Allowable Strength Design (ASD) for steel, the following special load combinations for seismic design shall be used as specifically required by Section 208, or by Chapter 3 through 7.

$1.2D + f_1L + 1.0E_m$ 203-19

$0.9D \pm 1.0E_m$ 203-20

Where

STRUCTURAL

$f_1 = 1.0$ for floors in places of public assembly, for live loads in excess of 4.8kPa, and for garage live load, or 0.5 for other live loads

E_m = the maximum effect of horizontal and vertical forces as set forth in Section 208.6.1

7. Dead load and superimposed dead load calculations

The dead load is the vertical load, which includes the weight of all permanent construction and all materials, and equipment, which are permanently fastened thereto and supported thereby.

Floor Load	2 nd -5 th Floor area load (kpa)
Vinyl Tiles 5 kg/m ²	0.05
Concrete fill finish (0.0023 kPa per mm)	1.15
Mechanical duct allowance	0.20
Suspended steel channel ceiling system	0.50
Partition load	1.00
Total	2.90

Floor Load	Roof Deck Area (kPa)
60 mm concrete floor topping sloping	1.42
Suspended steel channel ceiling system	0.50
Water proofing membrane	0.10
Total	2.00

LINE LOAD – SD (Superimposed dead)

Height of 150 mm thick CHB wall, h = floor to floor height minus beam depth

Unit weight of 150 mm thick CHB wall with plaster, $\gamma_c = 2.63 + 2 \times 0.24 = 3.11$ kPa

Line load for perimeter of building from GF to 5th floor = $\gamma_c \times h$ Glass line load from GF to 5th floor (*varies per floor*) = $(2.5 \times 9.81) \times (0.020) \times (h)$

External wall line load (kPa)

2nd floor = $3.050 \times 3.11 = 9.4855$

3rd floor = $2.750 \times 3.11 = 8.5525$

4th floor = $2.750 \times 3.11 = 8.5525$

5th floor = $2.750 \times 3.11 = 8.5525$

6th floor = $2.250 \times 3.11 = 6.9975$

Glass Line Load (kPa)

2nd floor = $2.5 \times 9.81 \times 3.050 \times 0.020 = 1.496025$

3rd floor = $2.5 \times 9.81 \times 2.750 \times 0.020 = 1.348875$

4th floor = $2.5 \times 9.81 \times 2.750 \times 0.020 = 1.348875$

5th floor = $2.5 \times 9.81 \times 2.750 \times 0.020 = 1.348875$

6th floor = $2.5 \times 9.81 \times 2.250 \times 0.020 = 1.103625$

Line Load at parapet perimeter = 3.73 kPa

8. Live Loads

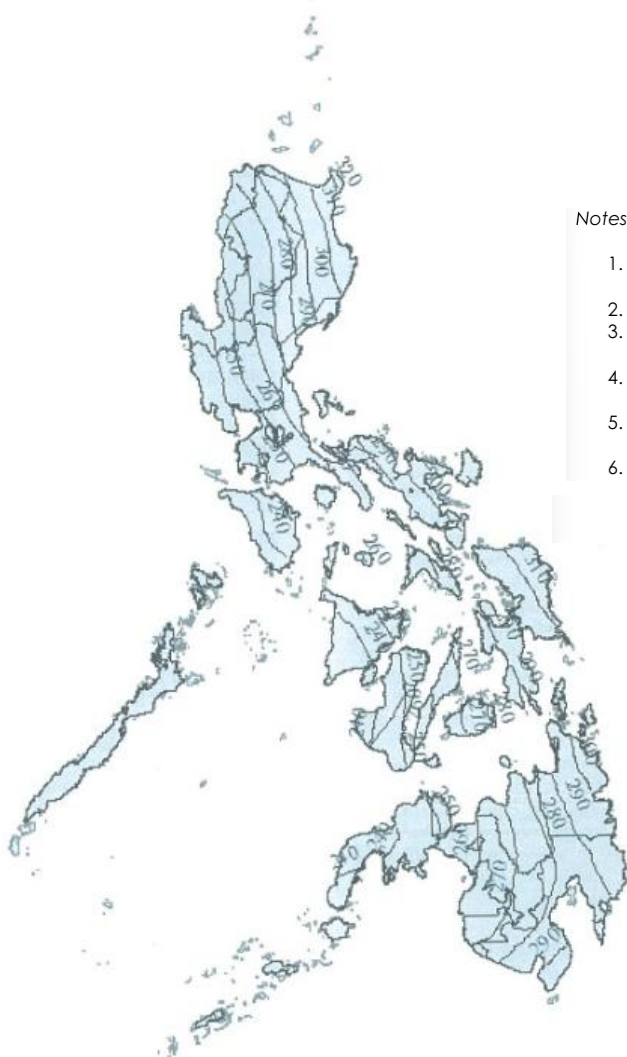
The live load includes loads that may vary in magnitude, and/or distribution during the life of the structure; not including wind load, earthquake or dead load. The minimum values of these loads depend on the occupancy of the structure and are normally specified by the governing codes. Table 2 shows the live loads used in the design of the building.

Live Loads

Live Loads	Load (kPa)
Office Use	2.4
Lift Lobby / Corridors	4.8
Roof Accessible	2.4
Stair Live Load	4.8

9. Wind Loads

Wind load is a force acting in the building, its building components and claddings that are exposed to the wind pressure intensity.



Notes:

1. Values are nominal design 3-second gust wind speeds in kilometres per hour at 10m above ground for exposure C category.
2. Linear interpolation between contours is permitted.
3. Islands and coastal areas outside the last contour shall use the last wind speed contour of the coastal area.
4. Mountainous terrain, gorges, ocean promontories, and special wind regions shall be examined for unusual wind conditions.
5. Wind speeds correspond to approximately a 15% probability of exceedance in 50 years (Annual Exceedance Probability = 0.00333, MRI = 300 years).
6. Results are from PAGASA.

Association of Structural Engineers of the Philippines, Inc. (ASEP)

Figure: Basic Wind Speeds for Occupancy Category III, IV and V Buildings and Other Structures (NSCP 2015 7th Edition)

Sample site location: **Upper General Luna Road, Baguio City**

Consider wind speed at 3 sec gust factor = 260 km/hr

Wind speed = 260 kilometer per hour

Wind speed input in ETABS = 161.59 mph

Exposure Category = **B**

Importance Factor = **1.00** (Special occupancy see NSCP 2015 Table 103-1 Occupancy)

Wind Load calculations based on NSCP 2015 adapted procedure form ASCE 7-10

Wind Load Procedure input in **ETABS = ASCE 7-16**

NSCP 2015 Directionality Factor, K_d

Table 207A.6-1
Wind Directionality Factor, K_d

Structure Type	Directionality Factor K_d *
Buildings	
Main Wind Force Resisting System	0.85
Components and Cladding	0.85
Arched Roofs	0.85
Chimneys, Tanks, and Similar Structures	
Square	0.90
Hexagonal	0.95
Round	0.95
Solid Freestanding Walls and Solid Freestanding and Attached Signs	0.85
Open Signs and Lattice Framework	0.85
Trussed Towers	
Triangular, square, rectangular	0.85
All other cross sections	0.95

**Directionality Factor K_d has been calibrated with combinations of loads specified in Section 203. This factor shall only be applied when used in conjunction with load combinations specified in Sections 203.3 and 203.4.*

Wind Load Parameters

NSCP 2015 Exposure Type

207A.7.2 Surface Roughness Categories

A ground Surface Roughness within each 45° sector shall be determined for a distance upwind of the site as defined in Section 207A.7.3 from the categories defined in the following text, for the purpose of assigning an exposure category as defined in Section 207A.7.3.

Surface Roughness B: Urban and suburban areas, wooded areas, or other terrain with numerous closely spaced obstructions having the size of single-family dwellings or larger.

Surface Roughness C: Open terrain with scattered obstructions having heights generally less than 9 m. This category includes flat open country and grasslands.

Surface Roughness D: Flat, unobstructed areas and water surfaces. This category includes smooth mud flats, salt flats, and unbroken ice.

207A.7.3 Exposure Categories

Exposure B: For buildings with a mean roof height of less than or equal to 9 m, Exposure B shall apply where the ground surface roughness, as defined by Surface Roughness B, prevails in the upwind direction for a distance greater than 450 m. For buildings with a mean roof height greater than 9 m, Exposure B shall apply where Surface Roughness B prevails in the upwind direction for a distance greater than 790 m or 20 times the height of the building, whichever is greater.

Exposure C: Exposure C shall apply for all cases where Exposures B or D do not apply.

Exposure D: Exposure D shall apply where the ground surface roughness, as defined by Surface Roughness D, prevails in the upwind direction for a distance greater than 1500 m or 20 times the building height, whichever is greater. Exposure D shall also apply where the ground surface roughness immediately upwind of the site is exposure B or C, and the site is within a distance of 180 m or 20 times the building height, whichever is greater, from an Exposure D condition as defined in the previous sentence.

Exposure B: For buildings with a mean roof height of less than or equal to 9 m, Exposure B shall apply where the ground surface roughness, as defined by Surface Roughness B, prevails in the upwind direction for a distance greater than 450 m. For buildings with a mean roof height greater than 9 m, Exposure B shall apply where Surface Roughness B prevails in the upwind direction for a distance greater than 790 m or 20 times the height of the building, whichever is greater.

For a site located in the transition zone between exposure categories, the category resulting in the largest wind forces shall be used.

NSCP 2015 Topographical Factor, K_{zt}

Table 207A.8-1
Topographic Multipliers for Exposure C

H/L_h	K_1 Multiplier			x/L_h	K_2 Multiplier		z/L_h	K_3 Multiplier		
	2-D Ridge	2-D Escarp.	3-D Axisym. Hill		2-D Escarp.	All Other Cases		2-D Ridge	2-D Escarp.	3-D Axisym. Hill
0.20	0.29	0.17	0.21	0.00	1.00	1.00	0.00	1.00	1.00	1.00
0.25	0.36	0.21	0.26	0.50	0.88	0.67	0.10	0.74	0.78	0.67
0.30	0.43	0.26	0.32	1.00	0.75	0.33	0.20	0.55	0.61	0.45
0.35	0.51	0.30	0.37	1.50	0.63	0.00	0.30	0.41	0.47	0.30
0.40	0.58	0.34	0.42	2.00	0.50	0.00	0.40	0.30	0.37	0.20
0.45	0.65	0.38	0.47	2.50	0.38	0.00	0.50	0.22	0.29	0.14
0.50	0.72	0.43	0.53	3.00	0.25	0.00	0.60	0.17	0.22	0.09
				3.50	0.13	0.00	0.70	0.12	0.17	0.06
				4.00	0.00	0.00	0.80	0.09	0.14	0.04
							0.90	0.07	0.11	0.03
							1.00	0.05	0.08	0.02
							1.50	0.01	0.02	0.00
							2.00	0.00	0.00	0.00

Notes:

- For values of H/L_h , x/L_h and z/L_h other than those shown, linear interpolation is permitted.
- For $\frac{H}{L_h} > 0.50$, assume $\frac{H}{L_h} = 0.50$ for evaluating K_1 and substitute $2H$ for L_h for evaluating K_2 and K_3 .
- Multipliers are based on the assumption that wind approaches the hill or escarpment along the direction of maximum slope.
- Notation:

Equations:

$$K_{zt} = (1 + K_1 + K_2 + K_3)^2$$

K_1 determined from table below

$$K_2 = \left(1 - \frac{|x|}{\mu L_h}\right)$$

$$K_3 = e^{-\gamma z/L_h}$$

207A.8.2 Topographic Factor

The wind speed-up effect shall be included in the calculation of design wind loads by using the factor K_{zt} :

$$K_{zt} = (1 + K_1 + K_2 + K_3)^2 \quad (207A.8-1)$$

where K_1 , K_2 , and K_3 are given in Figure 207A.8-1.

If site conditions and locations of structures do not meet all the conditions specified in Section 207A.8.1 then

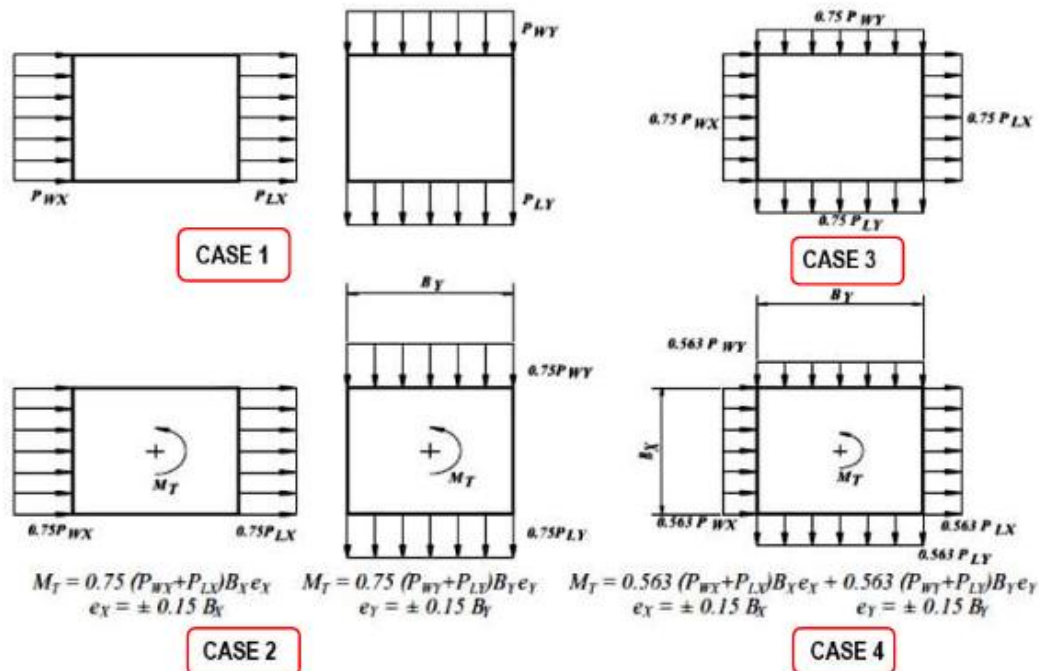
$$K_{zt} = 1.0.$$

NSCP 2015 Gust Effect Factor

207A.9.1 Gust Effect Factor

The gust-effect factor for a rigid building or other structure is permitted to be taken as 0.85.

NSCP 2015 / ASCE 7-10 Wind Load Cases



Case 1: Full design wind pressure acting on the projected area perpendicular to each principal axis of the structure, considered separately along each principal axis.

Case 2: Three quarters of the design wind pressure acting on the projected area perpendicular to each principal axis of the structure in conjunction with a torsional moment as shown, considered separately for each principal axis.

Case 3: Wind loading as defined in Case 1, but considered to act simultaneously at 75% of the specified value.

Case 4: Wind loading as defined in Case 2, but considered to act simultaneously at 75% of the specified value.

Notes:

- Design wind pressures for windward and leeward faces shall be determined in accordance with the provisions of 207B.4.1 and 207B.4.2 as applicable for building of all heights.
- Diagrams show plan views of building.
- Notation:

P_{WX}, P_{WY}	=	Windward face design pressure acting in the x, y principal axis, respectively.
P_{LX}, P_{LY}	=	Leeward face design pressure acting in the x, y principal axis, respectively.
$e(e_X, e_Y)$	=	Eccentricity for the x, y principal axis of the structure, respectively.
M_T	=	Torsional moment per unit height acting about a vertical axis of the building.

Figure 207B.4-8
Design Wind Load Cases All Heights

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10. Seismic Loads

NSCP 208.6.1

208.6 Earthquake Loads and Modeling Requirements

208.6.1 Earthquake Loads

Structures shall be designed for ground motion producing structural response and seismic forces in any horizontal direction. The following earthquake loads shall be used in the load combinations set forth in Section 203:

$$E = \rho E_h + E_v \quad (208-18)$$

$$E_m = \Omega_0 E_h \quad (208-19)$$

where:

- E = the earthquake load on an element of the structure resulting from the combination of the horizontal component, E_h , and the vertical component, E_v
- E_h = the earthquake load due to the base shear, V , as set forth in Section 208.5.2 or the design lateral force, F_p , as set forth in Section 208.9
- E_m = the estimated maximum earthquake force that can be developed in the structure as set forth in Section 208.6.1, and used in the design of specific elements of the structure, as specifically identified in this section
- E_v = the load effect resulting from the vertical component of the earthquake ground motion and is equal to an addition of $0.5C_v I D$ to the dead load effect, D , for Strength Design, and may be taken as zero for Allowable Stress Design
- Ω_0 = the seismic force amplification factor that is required to account for structural overstrength, as set forth in Section 208.4.10.1
- ρ = Reliability/Redundancy Factor as given by the following equation:

$$\rho = 2 - \frac{6.1}{r_{max}(\sqrt{A_B})} \quad (208-20)$$

where:

- r_{max} = the maximum element-storey shear ratio. For a given direction of loading, the element-storey shear ratio is the ratio of the design storey shear in the most heavily loaded single element divided by the total design storey shear.

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Seismic deadload or mass source

NSCP 208.6.1

Seismic dead load, W_s , is the total dead load and applicable portions of other loads listed below.

1. In storage and warehouse occupancies, a minimum of 25 percent of the floor live load shall be applicable.
2. Where a partition load is used in the floor design, a load of not less than 0.5 kN/m^2 shall be included.
3. Total weight of permanent equipment shall be included.

Seismic Active Fault Trace

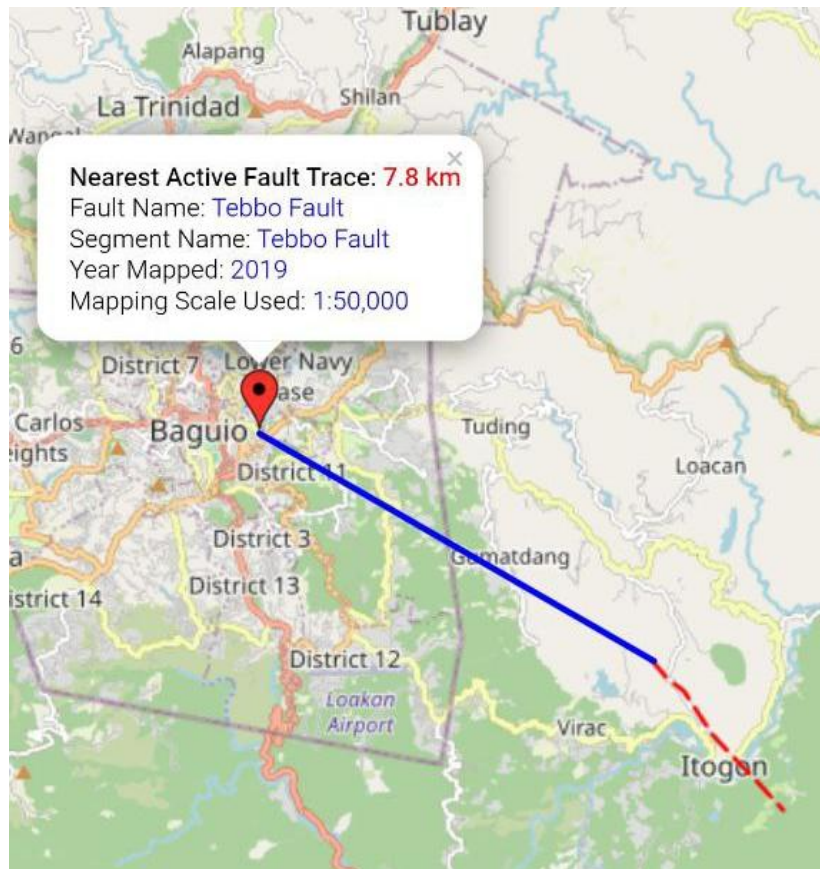


Figure: Location of the Nearest Fault from the Project Site (DOST-PHIVOLCS)

11. Occupancy Category

Standard occupancy category, **$I = 1.0$** FOR OFFICE BUILDING PROJECT
208.4.2 Occupancy Categories

For purposes of earthquake-resistant design, each structure shall be placed in one of the occupancy categories listed in Table 103-1. Table 208-1 assigns importance factors, I and I_p , and structural observation requirements for each category.

Table 208-1 - Seismic Importance Factors

Occupancy Category ¹	Seismic Importance Factor, I	Seismic Importance ² Factor, I_p
I. Essential Facilities ³	1.50	1.50
II. Hazardous Facilities	1.25	1.50
III. Special Occupancy Structures	1.00	1.00
IV. Standard Occupancy Structures	1.00	1.00
V. Miscellaneous structures	1.00	1.00

¹ See Table 103-1 for occupancy category listing

² The limitation of I_p for panel connections in Section 208.7.2.3 shall be **1.0** for the entire connector

³ Structural observation requirements are given in Section 107.9

⁴ For anchorage of machinery and equipment required for life-safety systems, the value of I_p shall be taken as **1.5**

12. Site Geology and Soil Characteristics

208.4.3 Site Geology and Soil Characteristics

Each site shall be assigned a soil profile type based on properly substantiated geotechnical data using the site categorization procedure set forth in Section 208.4.3.1.1 and Table 208-2.

Exception:

When the soil properties are not known in sufficient detail to determine the soil profile type, Type SD shall be used.

*Soil Profile Type **S_E** or **S_F** need not be assumed unless the building official determines that Type **S_E** or **S_F** may be present at the site or in the event that Type **S_E** or **S_F** is established by geotechnical data.*

208.4.3.1 Soil Profile Type

Soil Profile Types **S_A**, **S_B**, **S_C**, **S_D** and **S_E** are defined in Table 208-2 and Soil Profile Type **S_F** is defined as soils requiring site-specific evaluation as follows:

1. Soils vulnerable to potential failure or collapse under seismic loading, such as liquefiable soils, quick and highly sensitive clays, and collapsible weakly cemented soils.
2. Peats and/or highly organic clays, where the thickness of peat or highly organic clay exceeds 3.0 m.
3. Very high plasticity clays with a plasticity index, **PI** > **75**, where the depth of clay exceeds 7.5 m.
4. Very thick soft/medium stiff clays, where the depth of clay exceeds 35 m.
5. The criteria set forth in the definition for Soil Profile Type **S_F** requiring site-specific evaluation shall be considered. If the site corresponds to these criteria, the site shall be classified as Soil Profile Type **S_F** and a site-specific evaluation shall be conducted.

208.4.3.1.1 Site Categorization Procedure

208.4.3.1.1.1 Scope

This section describes the procedure for determining Soil Profile Types **S_A** through **S_F** as defined in Table 208-2.

Table 208-2 Soil Profile Types

Soil Profile Type	Soil Profile Name/ Generic Description	Average Soil Properties for Top 30 m of Soil Profile		
		Shear Wave Velocity, V_s (m/s)	SPT, N (blows/300 mm)	Undrained Shear Strength, S_u (kPa)
S_A	Hard Rock	> 1500		
S_B	Rock	760 to 1500		
S_C	Very Dense Soil and Soft Rock	360 to 760	> 50	> 100
S_D	Stiff Soil Profile	180 to 360	15 to 50	50 to 100
S_E^1	Soft Soil Profile	< 180	< 15	< 50
S_F	Soil Requiring Site-specific Evaluation. See Section 208.4.3.1			

13. Seismic Zone/Seismic Source type

208.4.4 Site Seismic Hazard Characteristics

Seismic hazard characteristics for the site shall be established based on the seismic zone and proximity of the site to active seismic sources, site soil profile characteristics and the structure's importance factor.

208.4.4.1 Seismic Zone

The Philippine archipelago is divided into two seismic zones only. Zone 2 covers the provinces of Palawan (except Busuanga), Sulu and Tawi-Tawi while the rest of the country is under Zone 4. Each structure shall be assigned a seismic zone factor Z, in accordance with Table 208-3.

Table 208-3 Seismic Zone Factor, Z

Zone	2	4
Z	0.20	0.40

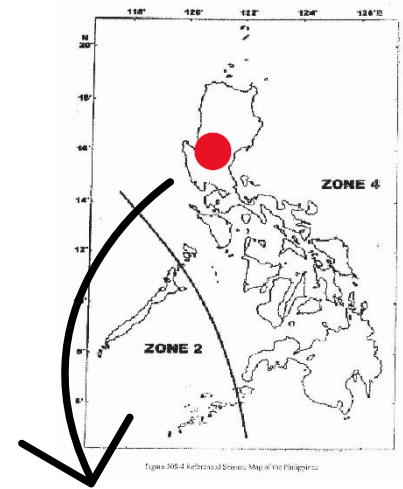
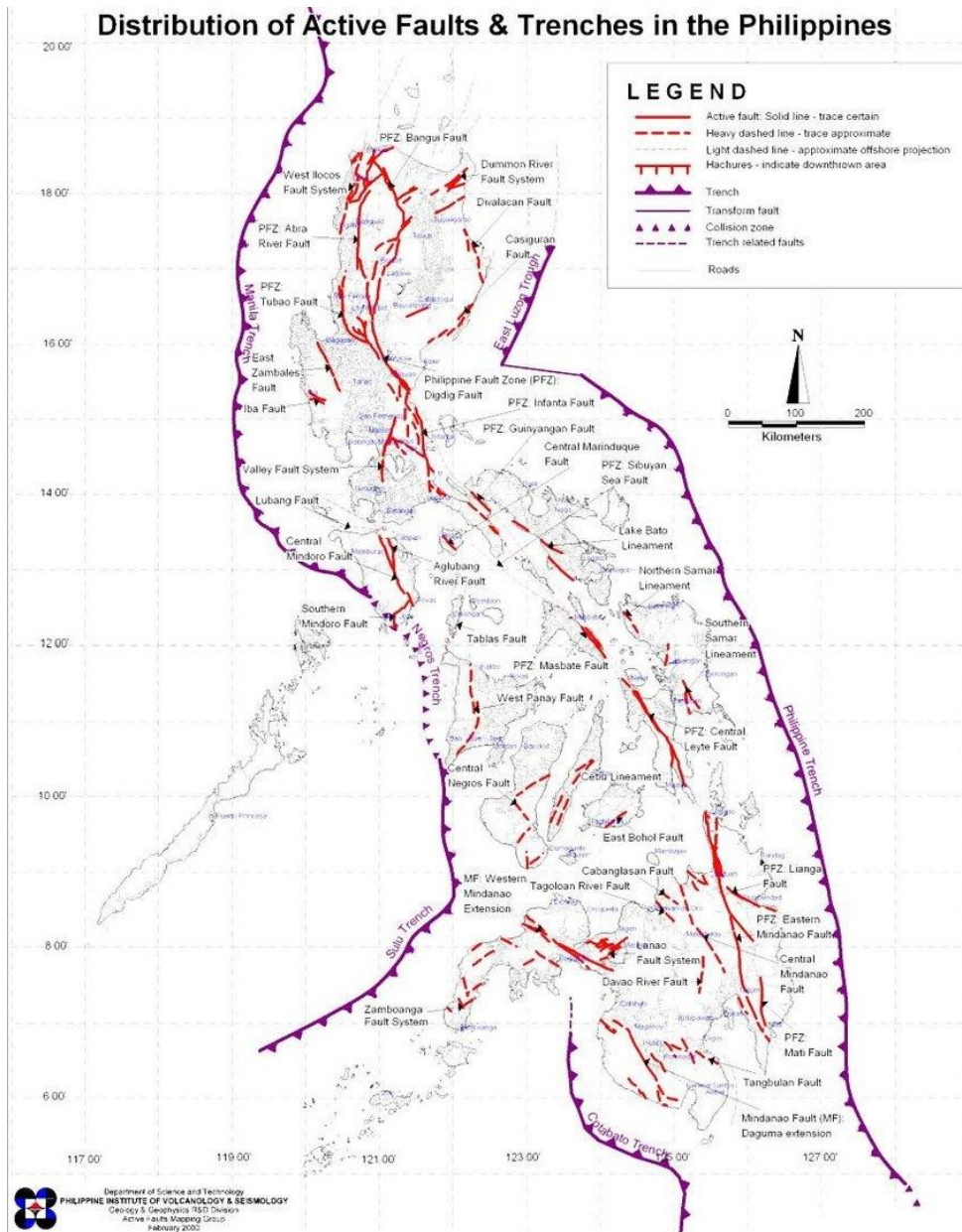
208.4.4.2 Seismic Source Types

Table 208-4 defines the types of seismic sources. The location and type of seismic sources to be used for design shall be established based on approved geological data. Type A sources shall be determined from Figure 208-2B, 2C, 2D, 2E or the most recent mapping of active faults by the Philippine Institute of Volcanology and Seismology (PHIVOLCS).

Section 208.4.4.2 Seismic Source Types

Table 208-4 Seismic Source Types

Seismic Source Type	Seismic Source Description	
A	Faults that are capable of producing large magnitude events and that have a high rate of seismic activity.	$7.0 \leq M \leq 8.4$
B	All faults other than Types A and C	$6.5 \leq M \leq 7.0$
C	Faults that are not capable of producing large magnitude earthquakes and that have a relatively low rate of seismic activity	$M < 6.5$



Zone 4

14. Seismic Zone 4 Near source and Response coefficients

Section 208.4.4.3 Seismic Zone 4 Near-Source Factor

Table 208-5 Near Source Factor, N_a

Seismic Source Type	Closest Distance To Known Seismic Source		
	≤ 2 km	≤ 5 km	≥ 10 km
A	1.5	1.2	1.0
B	1.3	1.0	1.0
C	1.0	1.0	1.0

Table 208-5 Near Source Factor, N_v

Seismic Source Type	Closest Distance To Known Seismic Source			
	≤ 2 km	≤ 5 km	≥ 10 km	≥ 15 km
A	2.0	1.6	1.2	1.0
B	1.6	1.2	1.0	1.0
C	1.0	1.0	1.0	1.0

Section 208.4.4.4 Seismic Response Coefficients

Table 208-7 Seismic Coefficient, C_a

Soil Profile Type	Closest Distance To Known Seismic Source	
	$Z = 0.2$	$Z = 0.4$
S_A	0.16	$0.32N_a$
S_B	0.20	$0.40N_a$
S_C	0.24	$0.40N_a$

S_D	0.28	$0.44N_a$
S_E^1	0.34	$0.44N_a$
S_F	See Footnote 1 of Table 208-8	

Table 208-8 Seismic Coefficient, C_v

Soil Profile Type	Closest Distance To Known Seismic Source	
	$Z = 0.2$	$Z = 0.4$
S_A	0.16	$0.32N_v$
S_B	0.20	$0.40N_v$
S_C	0.32	$0.56N_v$
S_D	0.40	$0.64N_v$
S_E^1	0.64	$0.96N_v$
S_F	See Footnote 1 of Table 208-8	

By interpolation to find N_a (Seismic source type A for 7.8Km)

$$\frac{N_{a_1} - N_a}{N_{a_1} - N_a} = \frac{d_1 - d_2}{d_1 - d_2} ; N_a = 1.088$$

By interpolation to find N_v (Seismic source type A for 7.8Km)

$$\frac{1.6 - N_v}{1.6 - 1.2} = \frac{5 - 7.8}{5 - 10} ; N_v = 1.376$$

$$C_a = 0.44 \times 1.088 = 0.47872$$

$$C_v = 0.48 \times 1.376 = 0.66048$$

15. Configuration Requirements

NSCP 208.4.5

208.4.5.1 Regular Structures

Regular structures have no significant physical discontinuities in plan or vertical configuration or in their lateral-force-resisting systems such as the irregular features described in Section 208.4.5.2.

208.4.5.2 Irregular Structures

1. Irregular structures have significant physical discontinuities in configuration or in their lateral- force-resisting systems. Irregular features include, but are not limited to, those described in Tables 208-9 and 208-10. All structures in occupancy Categories 4 and 5 in Seismic Zone 2 need to be evaluated only for vertical irregularities of Type 5 (Table 208-9) and horizontal irregularities of Type 1 (Table 208-10). 2. Structures having any of the features listed in Table 208-9 shall be designated as if having a vertical irregularity.

Exception:

Where no storey drift ratio under design lateral forces is greater than 1.3 times the storey drift ratio of the storey above, the structure may be deemed to not have the structural irregularities of Type 1 or 2 in Table 208-9. The storey drift ratio for the top two stories need not be considered. The storey drifts for this determination may be calculated neglecting torsional effects.

3. Structures having any of the features listed in Table 208-10 shall be designated as having a plan irregularity.

16. Vertical and horizontal Irregularities

Table 208-9 Vertical Structural Irregularities

Irregularity Type and Definition	Reference Section
1. Stiffness Irregularity - Soft Storey A soft storey is one in which the lateral stiffness is less than 70% of that in the storey above or less than 80 percent of the average stiffness of the three stories above.	208.4.8.3 Item 2
2. Weight (Mass) Irregularity Mass irregularity shall be considered to exist where the effective mass of any storey is more than 150% of the effective mass of an adjacent storey. A roof that is lighter than the floor below need not be considered.	208.4.8.3 Item 2
3. Vertical Geometric Irregularity Vertical geometric irregularity shall be considered to exist where the horizontal dimension of the lateral- force-resisting system in any storey is more than 130 % of that in an adjacent storey. One-storey penthouses need not be considered.	208.4.8.3 Item 2
4. In-Plane Discontinuity In Vertical Lateral-Force-Resisting Element Irregularity An in-plane offset of the lateral- load-resisting elements greater than the length of those elements.	208.5.8.1.5.1
5. Discontinuity In Capacity- Weak Storey Irregularity A weak storey is one in which the storey strength is less than 80% of that in the storey above. The storey strength is the total strength of all seismic-resisting elements sharing the storey for the direction under consideration.	208.4.9.1

Table 208-10 Horizontal Structural Irregularities

Irregularity Type and Definition	Reference Section
1. Torsional Irregularity - To Be Considered When Diaphragms Are Not Flexible Torsional irregularity shall be considered to exist when the maximum storey drift, computed including accidental torsion, at one end of the structure transverse to an axis is more than 1.2 times the average of the storey drifts of the two ends of the structure.	208.7.2.7 Item 6
2. Re-Entrant Corner Irregularity Plan configurations of a structure and its lateral-force-resisting system contain re-entrant corners, where both projections of the structure beyond a re-entrant corner are greater than 15 % of the plan dimension of the structure in the given direction.	208.7.2.7 Items 6 and 7
3. Diaphragm Discontinuity Irregularity Diaphragms with abrupt discontinuities or variations in stiffness, including those having cutout or open areas greater than 50% of the gross enclosed area of the diaphragm, or changes in effective diaphragm stiffness of more than 50 % from one storey to the next.	208.7.2.7 Item 6
4. Out-Of-Plane Offsets Irregularity Discontinuities in a lateral force path, such as out-of-plane offsets of the vertical elements	208.5.8.5. 1 208.7.2.7 Item 6;
5. Non-parallel Systems Irregularity The vertical lateral-load-resisting elements are not parallel to or symmetric about the major orthogonal axes of the lateral force-resisting systems.	208.7.1